

D465

Data Applications

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An analyst at a 60-store pet supply chain, which switched to a cloud data warehouse last quarter, needs the stores whose total 2025 sales exceeded \$2 million. The table sales(store_id, sale_date, amount) has 4.1 million rows. Which query is correct?

- A. SELECT store_id, SUM(amount) FROM sales WHERE sale_date BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY store_id HAVING SUM(amount) > 2000000;
- B. SELECT SUM(amount) FROM sales WHERE sale_date BETWEEN '2025-01-01' AND '2025-12-31' HAVING SUM(amount) > 2000000;
- C. SELECT store_id, SUM(amount) FROM sales WHERE SUM(amount) > 2000000 GROUP BY store_id;
- D. SELECT store_id, COUNT(amount) FROM sales WHERE sale_date BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY store_id HAVING COUNT(amount) > 2000000;

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

In a query that uses WHERE, GROUP BY, and HAVING, which statement is true?

- A. WHERE and HAVING are always interchangeable.
- B. WHERE filters rows before grouping; HAVING filters groups after aggregation.
- C. HAVING filters rows before grouping; WHERE filters groups after.
- D. HAVING runs before GROUP BY so the groups are smaller.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A nonprofit's donor database, maintained by a volunteer since 2019, has a donors table (2,000 rows, one per donor) and a gifts table (5,500 rows; 1,400 donors have given at least once). The director wants every donor with their gift total, including donors who have never given. Which join from donors to gifts should the analyst use?

- A. donors CROSS JOIN gifts
- B. donors RIGHT JOIN gifts
- C. donors LEFT JOIN gifts
- D. donors INNER JOIN gifts

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Using a donors table of 2,000 donors and a gifts table in which 1,400 distinct donors appear, how many donors does COUNT(DISTINCT donor_id) return after donors INNER JOIN gifts?

- A.** 600
- B.** 2,000
- C.** 5,500
- D.** 1,400

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A survey table at a hospital system with 14 clinics has 1,000 rows. The satisfaction column holds a score in 800 rows; 200 are NULL because patients skipped the question. The survey was designed by an outside firm. What does SELECT COUNT(satisfaction) FROM survey return?

- A.** An error, because the column contains NULLs.
- B.** 200
- C.** 1,000
- D.** 800

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A score column has five rows: 4, 5, NULL, 3, NULL. What does AVG(score) return?

- A.** NULL, because any NULL makes the result NULL.
- B.** 5
- C.** 4
- D.** 2.4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst at a regional airline whose fleet averages eleven years old has a data frame flights with 42 columns and 380,000 rows. She needs only the rows where delay_minutes is over 30. Which dplyr call does this?

- A. filter(flights, delay_minutes > 30)
- B. arrange(flights, desc(delay_minutes))
- C. filter(flights, delay_minutes >= 30)
- D. select(flights, delay_minutes > 30)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which dplyr call keeps all 380,000 rows of flights but only the columns carrier and delay_minutes?

- A. filter(flights, carrier, delay_minutes)
- B. select(flights, carrier, delay_minutes)
- C. mutate(flights, carrier, delay_minutes)
- D. summarize(flights, mean(delay_minutes))

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A budget analyst at a county with a \$310 million annual budget has department totals in B2:B20 and the grand total in B21. In C2 she wants each department's share of the total, and she will copy the formula down to C20. Her manager prefers percentages with one decimal. Which formula should go in C2?

- A. =B2/\$B\$21
- B. =SUM(B2:B20)/\$B\$21
- C. =B2/B21
- D. =\$B\$21/B2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A marketing analyst at an online bookstore that ships to 30 countries merges two customer lists and finds “Portland”, “portland ” and “PORTLAND” as separate cities, plus 312 exact duplicate rows. The campaign launches Monday. Before counting customers by city, what should she do?

- A.** Delete every row whose city is not in all caps.
- B.** Remove the duplicates only; case and spacing do not affect counts.
- C.** Count now and fix it after launch, since the date is fixed.
- D.** Trim spaces and standardize case, remove the exact duplicates, then count.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An analyst at a coffee roaster whose beans come from four countries writes `ggplot(sales, aes(x = month, y = revenue)) + geom_line(aes(color = 'blue'))`. The line comes out salmon-colored with a legend reading ‘blue’. Her team uses R 4.4. Why?

- A.** `geom_line` cannot take a color; use `geom_point` instead.
- B.** ‘blue’ is not a valid R color name.
- C.** The color was mapped inside `aes()` as a category; set it outside: `geom_line(color = 'blue')`.
- D.** R 4.4 changed the default color palette.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A health department analyst writes a weekly R Markdown report that pulls the latest case counts; the report goes to 11 county partners. Her director wants readers to see the charts and text but not the R code. What should she do?

- A.** Set `eval = FALSE` in the code chunks.
- B.** Set `echo = FALSE` in the code chunks.
- C.** Set `include = FALSE` in the code chunks.
- D.** Paste screenshots of the charts into a Word file each week.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

An analyst at a 60-store pet supply chain, ~~which switched to a cloud data warehouse last quarter,~~ needs the stores whose total 2025 sales exceeded \$2 million. The table sales(store_id, sale_date, amount) ~~has 4.1 million rows.~~ Which query is correct?

Target word: correct aggregate filter **Decides:** A condition on SUM(amount) per store belongs in HAVING.

A	Correct	WHERE limits rows to 2025; HAVING filters the per-store totals.
B	Dropped element	Drops GROUP BY, so it tests one company-wide total, not each store.
C	Adjacent category	WHERE runs before grouping and cannot contain an aggregate.
D	Confusion set	COUNT tallies transactions, not dollars.

2. Answer B Concept 1 • transfer

In a query that uses WHERE, GROUP BY, and HAVING, which statement is true?

Target word: order of filtering **Decides:** Rows are filtered, then grouped, then groups are filtered.

A	Absolute language	Only HAVING can test aggregates.
B	Correct	That is the logical processing order.
C	Inverted operation	Swaps the two clauses' roles.
D	Adjacent category	HAVING needs the groups to exist first.

3. Answer C Concept 2

A nonprofit's donor database, ~~maintained by a volunteer since 2019,~~ has a donors table (2,000 rows, one per donor) and a gifts table (5,500 rows; 1,400 donors have given at least once). The director wants every donor with their gift total, including donors who have never given. Which join from donors to gifts should the analyst use?

Target word: join type **Decides:** "Including donors who have never given" means keep all donors.

A	Confusion set	Pairs every donor with every gift: 11 million meaningless rows.
B	Inverted operation	Keeps every gift row, not every donor.
C	Correct	Keeps all 2,000 donors; non-givers get NULL totals (use COALESCE to show 0).
D	Dropped element	Drops the 600 donors with no gifts.

4. Answer D Concept 2 • transfer

Using a donors table of 2,000 donors and a gifts table in which 1,400 distinct donors appear, how many donors does COUNT(DISTINCT donor_id) return after donors INNER JOIN gifts?

Target word: result of an inner join **Decides:** INNER JOIN keeps only donors with a matching gift.

A	Inverted operation	The donors with no match, which the inner join removes.
B	Adjacent category	That is what a LEFT JOIN would keep.
C	Wrong stem number	The gift row count; DISTINCT counts donors.
D	Correct	Only donors who appear in gifts survive the inner join.

5. Answer D Concept 3

A survey table at a hospital system ~~with 14 clinics~~ has 1,000 rows. The satisfaction column holds a score in 800 rows; 200 are NULL because patients skipped the question. ~~The survey was designed by an outside firm.~~ What does SELECT COUNT(satisfaction) FROM survey return?

Target word: COUNT of a column with NULLs **Decides:** COUNT(column) skips NULLs.

A	Absolute language	Aggregates handle NULLs by skipping them.
B	Inverted operation	The NULL rows, which COUNT(column) ignores.
C	Confusion set	That is COUNT(*), which counts every row.
D	Correct	Only non-NULL scores are counted.

6. Answer C Concept 3 • transfer

A score column has five rows: 4, 5, NULL, 3, NULL. What does AVG(score) return?

Target word: AVG with NULLs **Decides:** AVG divides by the three non-NULL values.

A	Absolute language	That holds for arithmetic like $4 + \text{NULL}$, not for aggregates.
B	Adjacent category	That is MAX(score).
C	Correct	$(4 + 5 + 3) / 3 = 4$.
D	Confusion set	$12 / 5$ treats NULL as zero.

7. Answer A Concept 4

An analyst at a regional airline ~~whose fleet averages eleven years old~~ has a data frame flights ~~with 42 columns~~ and 380,000 rows. She needs only the rows where delay_minutes is over 30. Which dplyr call does this?

Target word: subset rows **Decides:** Keeping rows by a condition is filter().

A	Correct	filter() keeps rows meeting the condition.
B	Adjacent category	Sorts rows but keeps them all.
C	Wrong stem number	Also keeps flights delayed exactly 30 minutes, which are not over 30.
D	Confusion set	select() picks columns, not rows.

8. Answer B Concept 4 • transfer

Which dplyr call keeps all 380,000 rows of flights but only the columns carrier and delay_minutes?

Target word: subset columns **Decides:** Keeping columns is select().

A	Confusion set	filter() expects row conditions, not column names.
B	Correct	select() returns the named columns for every row.
C	Adjacent category	mutate() adds or changes columns and keeps the other 40.
D	Dropped element	Collapses all rows into one summary.

9. Answer A Concept 5

A budget analyst at a county ~~with a \$310 million annual budget~~ has department totals in B2:B20 and the grand total in B21. In C2 she wants each department's share of the total, and she will copy the formula down to C20. ~~Her manager prefers percentages with one decimal.~~ Which formula should go in C2?

Target word: formula that copies correctly **Decides:** The total cell must stay fixed when copied.

A	Correct	B2 shifts row by row; \$B\$21 stays anchored to the total.
B	Adjacent category	Returns 100% in every row instead of each department's share.
C	Dropped element	Without \$, the divisor shifts to B22, B23 ... when copied.
D	Inverted operation	Divides the total by the part.

10. Answer D Concept 6

A marketing analyst at an online bookstore ~~that ships to 30 countries~~ merges two customer lists and finds "Portland", "portland " and "PORTLAND" as separate cities, plus 312 exact duplicate rows. ~~The campaign launches Monday.~~ Before counting customers by city, what should she do?

Target word: cleaning step **Decides:** Two problems: inconsistent text and duplicate rows.

A	Absolute language	Throws away valid customers instead of standardizing them.
B	Dropped element	Variant spellings split one city into three groups.
C	Camouflage	The deadline does not make dirty counts correct.
D	Correct	Fixes both problems before aggregating.

11. Answer C Concept 7

An analyst at a coffee roaster ~~whose beans come from four countries~~ writes `ggplot(sales, aes(x = month, y = revenue)) + geom_line(aes(color = 'blue'))`. The line comes out salmon-colored with a legend reading 'blue'. ~~Her team uses R 4.4.~~ Why?

Target word: cause of the wrong color **Decides:** A constant was mapped inside `aes()`.

A	Absolute language	<code>geom_line</code> accepts color.
B	Adjacent category	It is valid; the problem is where it was placed.
C	Correct	Inside <code>aes()</code> maps data; a fixed color is set outside <code>aes()</code> .
D	Camouflage	The version is irrelevant; the mapping causes the legend.

12. Answer B Concept 8

A health department analyst writes a weekly R Markdown report that pulls the latest case counts; ~~the report goes to 11 county partners.~~ Her director wants readers to see the charts and text but not the R code. What should she do?

Target word: chunk option **Decides:** Code must run and show output, but the code is hidden.

A	Confusion set	The code would not run, so no charts would appear.
B	Correct	Code runs and output shows; the code itself is hidden.
C	Dropped element	Hides the output too, so the charts disappear.
D	Adjacent category	Loses the reproducible, auto-updating report.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	WHERE vs. HAVING	WHERE filters rows before grouping; HAVING filters groups after aggregation.	Putting an aggregate like SUM() in WHERE.	A condition on a total, count, or average per group.
2	INNER vs. LEFT JOIN	INNER JOIN keeps only matching rows; LEFT JOIN keeps every row of the left table, with NULLs where no match exists.	Using INNER JOIN when the request says “including those with none.”	“Every customer, even those who never ...”
3	NULLs in aggregates	COUNT(column) and AVG() skip NULLs; COUNT(*) counts all rows.	Treating NULL as zero.	A column with missing values fed to COUNT or AVG.
4	filter() vs. select() in dplyr	filter() keeps rows that meet a condition; select() keeps columns.	Using select() to subset rows.	“Only rows where ...” vs. “only the columns ...”
5	Absolute cell references	A \$ anchors a row or column so it does not shift when a formula is copied.	Copying a formula that divides by a relative reference to a total.	A formula copied down that refers to one fixed total cell.
6	Cleaning before counting	Standardize text (trim, case) and remove duplicates before aggregating.	Fixing only one kind of problem or counting first.	The same value spelled several ways plus duplicate rows.
7	ggplot2 mapping vs. setting	Inside aes() maps a variable to an aesthetic; a fixed value such as a color goes outside aes().	Putting a constant color inside aes().	A legend shows a color name as a category.
8	R Markdown chunk options	echo = FALSE hides code but shows output; eval = FALSE skips running; include = FALSE runs but hides everything.	Using eval = FALSE to hide code.	“Readers should see results but not code.”

Course scope (public description): Data Applications examines the principles and techniques used to effectively analyze data to answer questions and provides foundational knowledge of R and RStudio environments. This course teaches students how to aggregate; how to format and adjust data using spreadsheets and SQL; how to use formulas and functions to perform calculations in SQL; how to organize, transform, clean, and analyze data in R; and how to create visualizations in R Markdown. Upon completion of this course, students will earn the WGU Business Analysis Professional certificate and may also choose to earn the Google Data Analytics Professional certificate.